

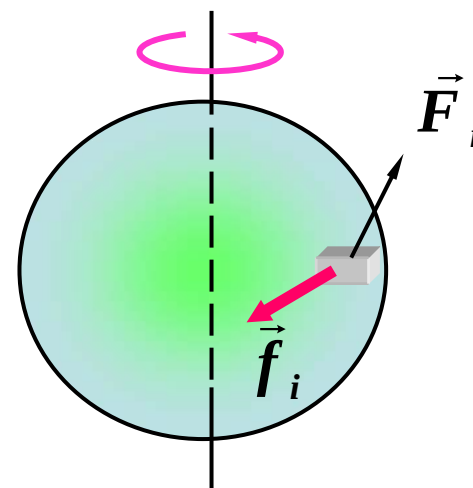
§ 2.2 刚体的定轴转动定律及其应用

2.2.1 刚体的定轴转动定律

根据质点系的角动量定理，
对于刚体

$$\vec{M}_{\text{外}} = \frac{d\vec{L}}{dt}$$

作用于刚体上的合外力矩等于
刚体角动量对时间的变化率



对于定轴转动的刚体，沿转轴 (z 轴) 方向得

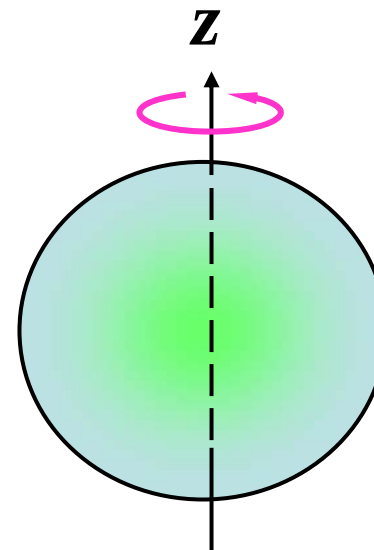
$$\vec{M}_{\text{外}} = \frac{d\vec{L}}{dt}$$

$$M_z = \frac{dL_z}{dt}$$

$$L_z = J\omega$$

刚体的定轴转动定律

$$M_{\text{外}} = J\beta$$





讨论

刚体定轴转动定律 $M_{\text{外}} = J\beta$

1. $M \rightarrow F, \beta \rightarrow a, J \rightarrow m \quad \vec{F} = m \vec{a}$

M 相同, J 越大, β 越小,

J 反映了定轴转动刚体的惯性。

2. $\beta \propto M$, 两者方向相同

3. 转动第一定律 若 $M = 0, \beta = 0$

刚体保持原有的角速度不变

静止的仍静止, 转动的保持角速度不变

关于力矩沿转轴分量的讨论

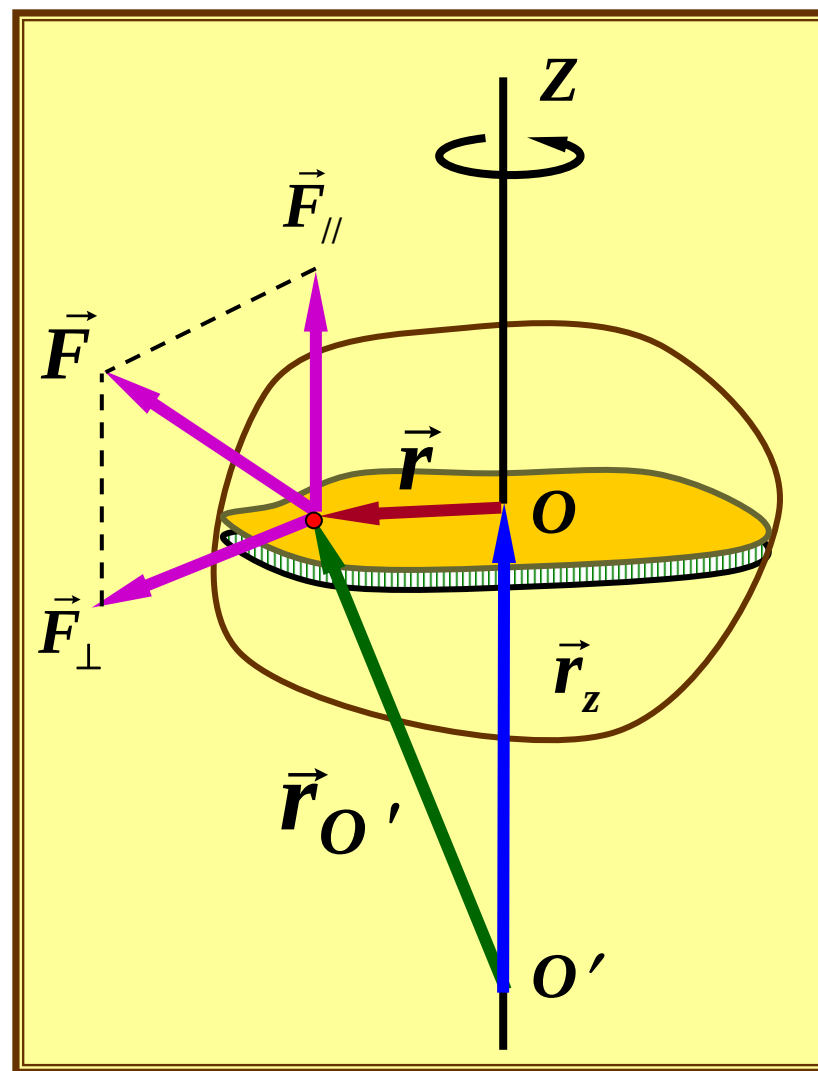
$$\vec{M}_{O'} = \vec{r}_{O'} \times \vec{F} = \vec{r}_{O'} \times (\vec{F}_{\perp} + \vec{F}_{\parallel})$$

$$= \vec{r}_{O'} \times \vec{F}_{\perp} + \vec{r}_{O'} \times \vec{F}_{\parallel}$$

$$= (\vec{r} + \vec{r}_z) \times \vec{F}_{\perp} + \vec{r}_{O'} \times \vec{F}_{\parallel}$$

$$= \vec{r} \times \vec{F}_{\perp} + \underbrace{\vec{r}_z \times \vec{F}_{\perp} + \vec{r}_{O'} \times \vec{F}_{\parallel}}_{\text{垂直于转轴}}$$

$$\vec{M}_z = \vec{r} \times \vec{F}_{\perp}$$



结论：对于定轴转动的刚体

若外力在垂直于转轴的平面内

$$\vec{M} = \vec{r} \times \vec{F}$$

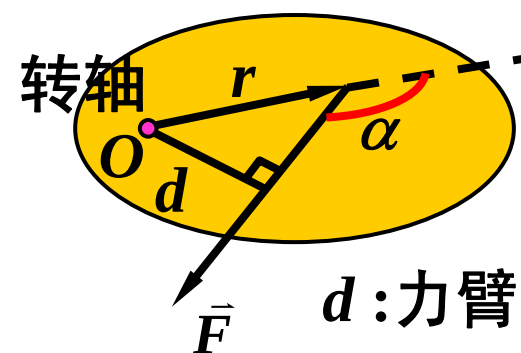
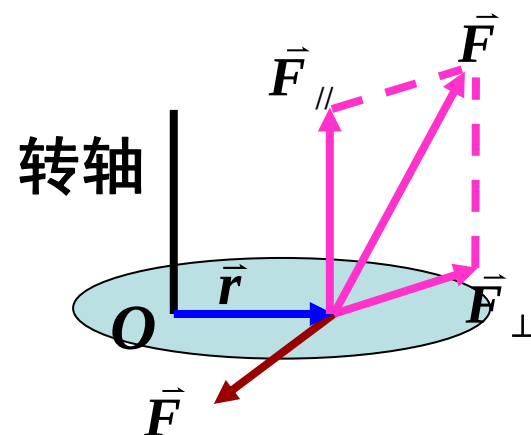
$$M = rF \sin \alpha = Fd$$

若外力不在垂直于转轴的平面内

$$\vec{F} = \vec{F}_{\perp} + \vec{F}_{\parallel}$$

$\vec{F}_{\parallel}$ 平行于转轴，不会使刚体绕轴转动
对转轴的力矩为零

$$\vec{M} = \vec{r} \times \vec{F}_{\perp}$$



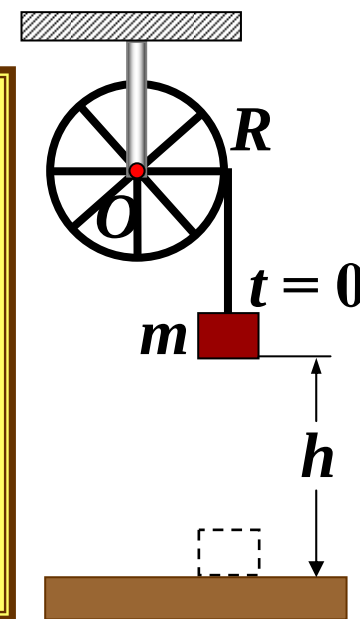
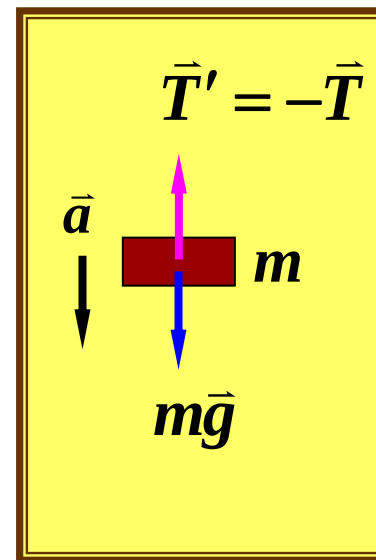
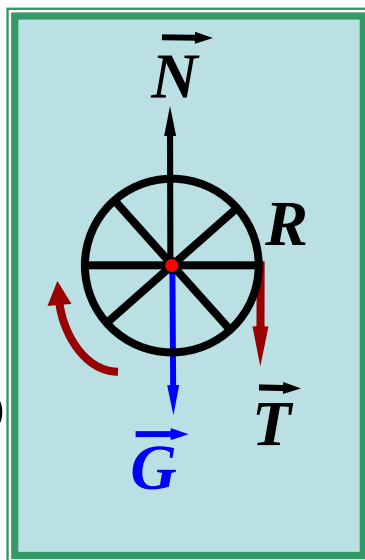
2.2.2 转动定律应用举例

例：已知：轮 $R = 0.2\text{m}$ ，物体质量 $m = 1\text{kg}$ ， $v_0 = 0$ ， $h = 1.5\text{m}$ ，绳子与轮子间无相对滑动，绳子不可伸长，下落时间 $t = 3\text{s}$ 。

求：轮对 O 轴的转动惯量 J 。

解：对轮应用转动定律

$$\begin{cases} T R = J \beta \\ mg - T = ma \\ a = R\beta \text{ (无相对滑动且绳子不伸长)} \\ h = \frac{1}{2}at^2 \end{cases}$$



解得 $J = \left(\frac{gt^2}{2h} - 1\right)mR^2 = \left(\frac{9.8 \times 3^2}{2 \times 1.5} - 1\right) \times 1 \times 0.2^2 = 1.14 \text{ (kg} \cdot \text{m}^2\text{)}$

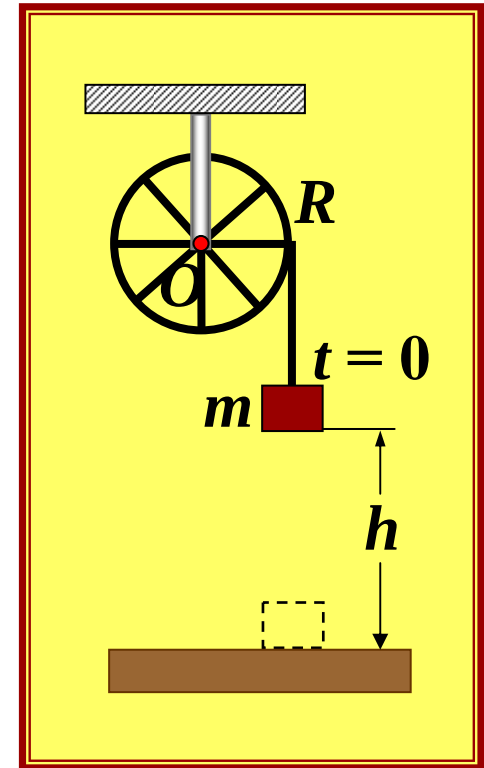
分析: 1. h 、 m 一定, $J \uparrow \rightarrow t \uparrow$, 合理

2. 物体的运动

$$T = \frac{J}{J + mR^2} mg \quad a = \frac{mR^2}{J + mR^2} g$$

3. 若 $J = 0$, 得 $h = \frac{1}{2} gt^2$, 正确

4. 若 $J \gg mR^2$, $T \approx mg$ 且 $a \approx 0$



例：长度为 l ，质量为 m 的均匀细棒，在竖直平面内摆动。棒最初处于水平位置，求它下摆到 θ 角时的角加速度和角速度。

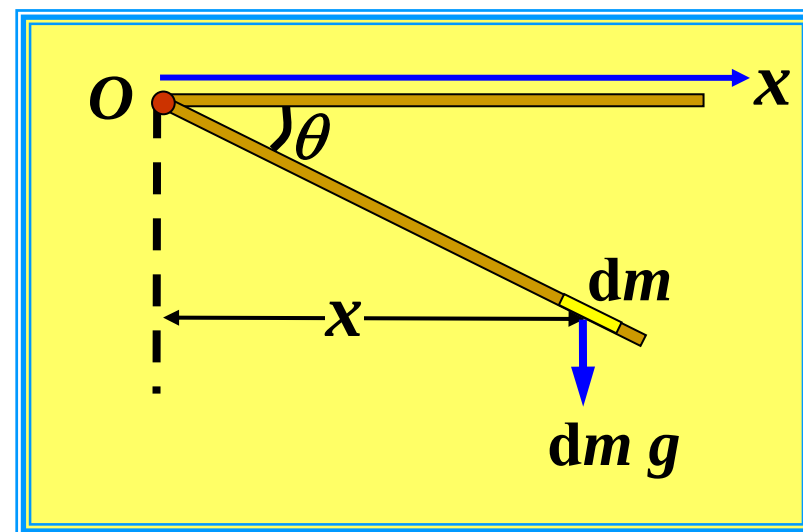
解：先求角加速度

$$dM = x dm g \quad \text{方向: } \otimes$$

$$M = \int x g dm = mg x_c = \frac{1}{2} mgl \cos \theta$$

由转动定律 $M = J\beta$ $J = \frac{1}{3} ml^2$

$$\beta = \frac{M}{J} = \frac{3g \cos \theta}{2l}$$



$$x_c = \frac{\int x dm}{m}$$

求角速度

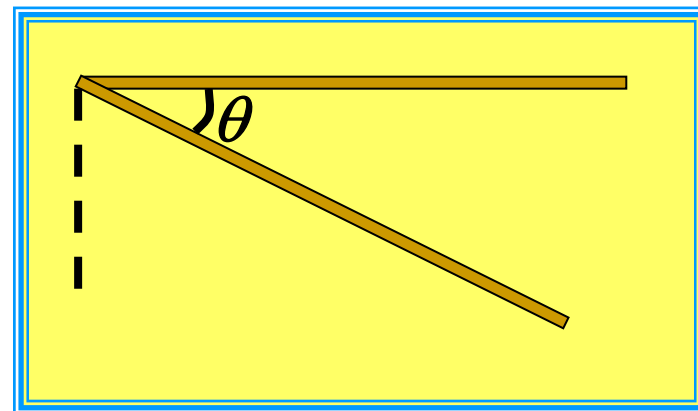
$$\beta = \frac{d\omega}{dt} = \frac{d\omega}{d\theta} \frac{d\theta}{dt} = \omega \frac{d\omega}{d\theta}$$

$$\omega d\omega = \beta d\theta$$

$$\int_0^{\omega} \omega d\omega = \int_0^{\theta} \beta d\theta$$

$$\frac{1}{2} \omega^2 = \int_0^{\theta} \frac{3g \cos \theta}{2l} d\theta$$

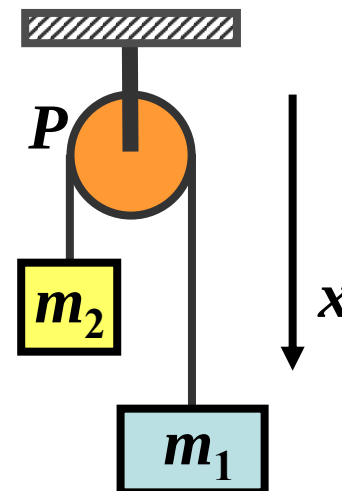
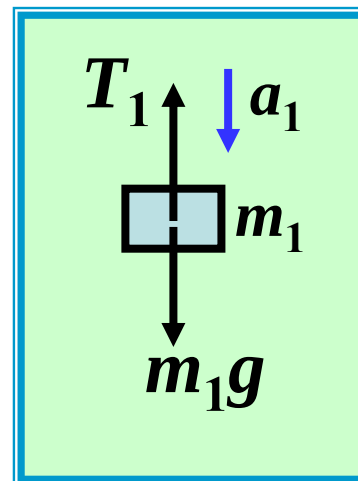
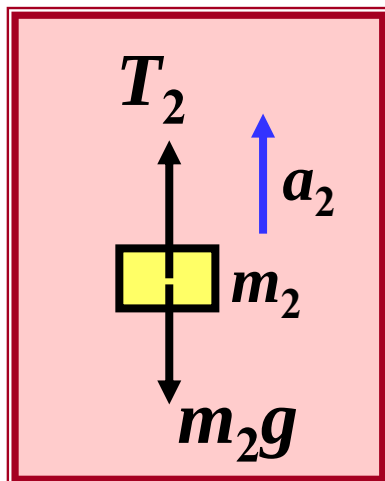
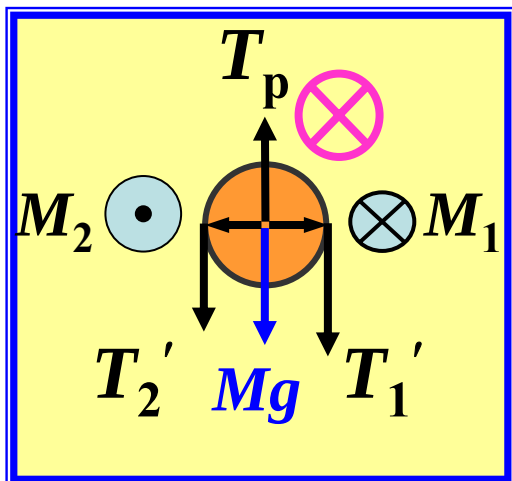
$$\omega = \sqrt{\frac{3g \sin \theta}{l}}$$



$$\beta = \frac{3g \cos \theta}{2l}$$

例：轻绳下挂一半径为 R 、质量为 M 的滑轮 P 。 P 上跨一长度变化可忽略的轻绳，绳两端挂 m_1 、 m_2 ($m_1 > m_2$)。求： m_1 、 m_2 的加速度，两侧绳中的张力（绳子和滑轮间不打滑）。

解：



$$T_1 R - T_2 R = J\beta \quad T_2 - m_2 g = m_2 a \quad m_1 g - T_1 = m_1 a$$

$$J = \frac{1}{2} M R^2$$

$$a_1 = a_2 = a \quad (\text{绳子不可伸长})$$

$$\left\{ \begin{array}{l} m_1 g - T_1 = m_1 a \\ T_2 - m_2 g = m_2 a \\ T_1 R - T_2 R = J \beta \\ a = R \beta \\ J = \frac{1}{2} M R^2 \end{array} \right.$$

$$\left\{ \begin{array}{l} a = \frac{(m_1 - m_2)g}{(m_1 + m_2 + \frac{1}{2}M)} \\ T_1 = \frac{m_1(2m_2 + \frac{1}{2}M)g}{(m_1 + m_2 + \frac{1}{2}M)} \\ T_2 = \frac{m_2(2m_1 + \frac{1}{2}M)g}{(m_1 + m_2 + \frac{1}{2}M)} \\ \beta = \frac{(m_1 - m_2)g}{(m_1 + m_2 + \frac{1}{2}M)R} \end{array} \right.$$

